

organisation stands in relation to IPv6. If your website is hosted by an Internet Service Provider (ISP), ask its customer support personnel the following questions:

- Do you currently provide IPv6 connectivity?
- If not, when do you plan to deploy IPv6 on your customer networks? What is your deployment time-line?
- When will my website be available over IPv6? What extra cost will that entail?
- Do you provide customers with IPv6-compatible modems, or other devices necessary for connecting over IPv6?
- Why have you not provided us any information on IPv6 that would help us put in place all that's needed from our side, to be ready for IPv6 compliance?

Next, carry out an audit of your entire IT infrastructure, and identify the hardware/software that needs to be changed or upgraded. For your hardware (routers, servers, etc), get in touch with the OEM or vendor to check for IPv6 compliance. Ensure that all the equipment is dual stacked (running both IPv4 and IPv6). If the hardware does not support IPv6, find out from the OEM whether a replacement is required, or if an upgrade/patch will suffice.

For software purchased from a third party, get in touch with the provider to check if the product is already IPv6 compatible, or if there's an upgrade available. If no IPv6 upgrade is available, look for an alternative software source. Any software that has been developed in-house may have to be rewritten. Identify and change all hard-coded IPv4 addressees in software code to IPv6. Any application that directly uses network sockets or other network-related programming might have to be re-written. Most operating systems are IPv6 compatible—IPv6 support is built in, and you only need to activate it. However, identify any legacy systems running an OS without IPv6 support, and either replace them, or patch/upgrade the OS.

I use FOSS; am I affected?

IPv6 migration is an issue that affects everyone. It is not restricted to only particular hardware or software. The problem is independent of the technology or software you use, as it stems from a core flaw in the TCP/IP addressing scheme. So yes, people using open source operating systems and applications could also be affected.

Linux and IPv6

The first IPv6-related network code was added to the Linux kernel 2.1.8 in November 1996 by Pedro Roque. In most Linux configurations these days, IPv6 is active by default, and the required IPv6 kernel modules are loaded automatically. To check whether your current running kernel supports IPv6, look in your `/proc` file-system for the pseudo-file `/proc/net/ipv6`. If this entry is not present, you can try to load the IPv6 module by executing `modprobe ipv6`. If successful, this module should be listed by the following command:

```
# lsmod | grep -w 'ipv6' && echo "IPv6 module successfully loaded"
```

It is also possible to automatically load the IPv6 module on demand. You have to add the following line to the kernel module-loader configuration file, which is normally `/etc/modules.conf` or `/etc/conf.modules`:

```
alias net-pf-10 ipv6 # automatically load IPv6 module on demand
```

If your kernel has no IPv6 support, then you can try one of the following options:

- Update your Linux distribution to a current one that supports IPv6 out-of-the-box.
- Compile a new vanilla kernel.
- Recompile kernel sources given by your Linux distribution.
- Compile a kernel with third-party IPv6 extensions.

However, it is pertinent to point out that IPv6 support in the kernel and the other basic packages is pretty average in terms of performance, and does not conform to all drafts and RFCs (request for comments). Most people tend to switch off the IPv6-compliance feature, as it makes the system quite slow.

To overcome this, you can make use of the USAGI (Universal Playground for IPv6) kit, which contains an enhanced IPv6-enabled version of both the latest 2.6.x Linux kernel, and the most important networking software packages for Linux. USAGI is a project run by volunteers from various organisations, and aims to develop a production-quality IPv6 protocol stack and IPsec for Linux. Based in Japan, USAGI teams up with other such projects like the WIDE Project, the KAME Project and TAHI, to implement all absent or obsolete IPv6 support in Linux. The latest release of the USAGI kit can be downloaded from <http://www.linux-ipv6.org/>.

Apache and IPv6

The Apache HTTP server supports IPv6 on most platforms, allowing Apache to allocate IPv6 sockets and handle requests sent over IPv6. However, to allow the IPv6 socket to handle both IPv4 and IPv6 connections, you will need to tweak the configuration parameter and use generic Listen directives. To verify that Apache is working in dual-stack mode (supporting both IPv4 and IPv6), use the `netstat` command.

The writing is on the wall. With IPv4 addresses having been exhausted, the time, cost and complexity of operating a stable and growing Internet will rise significantly, unless IPv6 deployment is well under way. So hurry and make your organisation IPv6-ready, now! 

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